

A. Feder Cooper

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I research a variety of topics in reliable, scalable machine learning. My contributions span privacy, security, evaluation of generative-AI systems, MLSys, and measurement validity. I also do work in tech policy and law, and spend a lot of time finding ways to effectively communicate the capabilities and limits of AI/ML to interdisciplinary audiences and the public. My research has received spotlights, orals, and best-paper accolades at top AI/ML and computing venues, including *NeurIPS*, *ICML*, *AAAI*, and *AIES*. Collaborations on copyright and generative AI have been lauded as “landmark” work among scholars and the international press.

Current Position

Assistant Professor, Department of Computer Science, Yale University

Co-lead, The GenLaw Center

Affiliate Researcher, Stanford HAI, RegLab, and CRFM

Faculty Associate at the Berkman Klein Center for Internet & Society at Harvard University

Education

Cornell University

Ph.D., Computer Science

2018 – 2024

M.S., Computer Science

2021

Columbia University

B.A., Computer Science and Archaeology, Phi Beta Kappa, *summa cum laude*

2014

Selected Papers

*Denotes co-first author or equal contribution; †lead author (not specified if same as first author); [Google Scholar](#)

A. Feder Cooper, Marika Swanberg, Jamie Hayes, Lea Duesterwald, Christopher De Sa, Daniel E. Ho, Mark A. Lemley, and Percy Liang. “Extractable Memorization From First Principles.” 2026. [[link](#)] [Preprint](#)

Mark A. Lemley* and **A. Feder Cooper***. “Probabilistic ‘Copies’ in Generative AI Models.” Forthcoming, *Berkeley Technology Law Journal*. 2026. [[link](#)] [Journal](#)

A. Feder Cooper, Mark A. Lemley, Christopher De Sa, Daniel E. Ho, Percy Liang, and others. “Extracting memorized pieces of (copyrighted) books from open-weight language models.” *COLM '26*. [[link](#)] [Proceedings](#) (29% acceptance; written 2025; *ICML Workshop* [Oral](#))

A. Feder Cooper, Mark A. Lemley, Christopher De Sa, Lea Duesterwald, Allison Casasola, Jamie Hayes, Katherine Lee, Daniel E. Ho, and Percy Liang. “Estimating near-verbatim extraction risk in language models with decoding-constrained beam search.” *COLM '26*. [[link](#)] [Proceedings](#) (29% acceptance; *ICML Workshop* [Oral](#))

Miles Brundage, Noemi Dreksler, Aidan Homewood, Sean McGregor, Patricia Paskov, Conrad Stosz, Girish Sastri, **A. Feder Cooper**, and others. “Frontier AI Auditing: Toward Rigorous Third-Party Assessment of Safety and Security Practices at Leading AI Companies.” 2026. [[link](#)] [Whitepaper](#)

Ahmed Ahmed*, **A. Feder Cooper***, Sanmi Koyejo, and Percy Liang. “Extracting books from production language models.” 2026. [[link](#)] [Preprint](#)

Jamie Hayes*, **A. Feder Cooper***, Ilia Shumailov, Christopher A. Choquette-Choo, Matthew Jagielski, Milad Nasr, Katherine Lee, and others. “Exploring the limits of strong membership inference attacks on large language models.” *NeurIPS '25*. [[link](#)] [Proceedings](#) (24.52% acceptance)

A. Feder Cooper^{*†}, Katherine Lee^{*†}, Miranda Bogen*, James Grimmelmänn, danah boyd, Percy Liang, Daniel E. Ho, Pamela Samuelson, Amy B. Cyphert, Mark A. Lemley, Nicolas Papernot, and others. “Machine Unlearning Doesn’t Do What You Think: Lessons for Generative AI Policy and Research.” *NeurIPS ’25, Position Track*. [[link](#)] **Proceedings** **Oral** (≈0.4% of submitted papers).

Jamie Hayes*, Marika Swanberg, Harsh Chaudhari, Itay Yona, Ilia Shumailov, Milad Nasr, Christopher A. Choquette-Choo, Katherine Lee, and **A. Feder Cooper***. “Measuring memorization in language models via probabilistic extraction.” *NAACL ’25*. [[link](#)] **Proceedings** (≈22% acceptance)

A. Feder Cooper* and James Grimmelmänn*. “The Files are in the Computer: On Copyright, Memorization, and Generative AI.” *Chicago-Kent Law Review*, Vol. 100, 2025. [[link](#)] **Journal** (written 2024)

Katherine Lee*, **A. Feder Cooper***, and James Grimmelmänn*. “Talkin’ ’Bout AI Generation: Copyright and the Generative-AI Supply Chain.” *Journal of the Copyright Society*, Volume 72, 2025. [[link](#)] **Journal** (written 2023)

Nicholas Carlini, Daniel Paleka, Krishnamurthy Dj Dvijotham, Thomas Steinke, Jonathan Hayase, **A. Feder Cooper**, Katherine Lee, Matthew Jagielski, Milad Nasr, Arthur Conmy, Eric Wallace, David Rolnick, and Florian Tramèr. “Stealing Part of a Production Language Model.” *ICML ’24*. [[link](#)] **Proceedings** **Best Paper** (≈0.1% of submitted papers)

A. Feder Cooper, Solon Barocas, Christopher De Sa, Jon Kleinberg, and others. “Arbitrariness and Social Prediction: The Confounding Role of Variance in Fair Classification.” *AAAI ’24*. [[link](#)] **Proceedings** **Best Student Paper Honorable Mention** (<1% of submitted papers; written 2023)

A. Feder Cooper^{*†}, Wentao Guo*, Khiem Pham*, and others. “Coordinating Distributed Example Orders for Provably Accelerated Training.” *NeurIPS ’23*. [[link](#)] **Proceedings** (26% acceptance)

Milad Nasr*, Nicholas Carlini*, Jonathan Hayase, Matthew Jagielski, **A. Feder Cooper**, Daphne Ippolito, Christopher A. Choquette-Choo, Eric Wallace, Florian Tramèr, and Katherine Lee[†]. “Scalable Extraction of Training Data from (Production) Language Models.” 2023. [[link](#)] **Tech Report**

A. Feder Cooper^{*†}, Katherine Lee*, James Grimmelmänn*, Daphne Ippolito*, Miles Brundage, Jack M. Balkin, Deep Ganguli, Mark Lemley, Corynne McSherry, Pamela Samuelson, Luis Villa, and others. “Report of the 1st Workshop on Generative AI and Law.” 2023. [[link](#)] **Tech Report**

A. Feder Cooper, Jonathan Frankle, and Christopher De Sa. “Non-Determinism and the Lawlessness of Machine Learning Code.” *CSLAW ’22*. [[link](#)] **Proceedings**

A. Feder Cooper^{*†}, Emanuel Moss*, Benjamin Laufer, and Helen Nissenbaum[†]. “Accountability in an Algorithmic Society: Relationality, Responsibility, and Robustness in Machine Learning.” *FAccT ’22*. [[link](#)] **Proceedings** (25% acceptance)

A. Feder Cooper, Yucheng Lu, Jessica Zosa Forde, and Christopher De Sa. “Hyperparameter Optimization Is Deceiving Us, and How to Stop It.” *NeurIPS ’21*. [[link](#)] **Proceedings** (<26% acceptance)

A. Feder Cooper and Ellen Abrams. “Emergent Unfairness in Algorithmic Fairness-Accuracy Trade-Off Research.” *AIES ’21*. [[link](#)] **Proceedings** **Oral** (< 10% of submitted papers)

Ruqi Zhang*, **A. Feder Cooper***, and Christopher De Sa[†]. “Asymptotically Optimal Exact Minibatch Metropolis-Hastings.” *NeurIPS ’20*. [[link](#)] **Proceedings** **Spotlight** (<3% of submitted papers)

Selected Honors, Grants, and Awards

Stanford HAI Seed Grant (\$75,000), “Assessing Copyright Risks of Training Data ‘Memorization’ and ‘Extraction’ for Open-Weight Language Models” (2026 – 2027)

Meta Ph.D. Fellowship Finalist (2023) (short-listed out of 3200 candidates)

Rising Star in EECS, MIT (2021)

Fellowship Finalist, OpenPhil AI Ph.D. Fellowship (2021)

Selected Conference, Workshop, and Tutorial Organization

Data Privacy, Memorization, & Legal Implications in Generative AI: A Practical Guide. *NeurIPS '25* Tutorial, December 2, 2025 (San Diego, CA).

2nd Workshop on Generative AI and Law (GenLaw '24). Held at *ICML 2024* in Vienna, Austria. Lead co-organizer with Katherine Lee. [[link](#)]

Evaluating Generative AI Systems: the Good, the Bad, and the Hype (GenLaw DC '24). Lead co-organizer. Sponsored by the K&L Gates initiative in Ethics and Computational Technologies at Carnegie Mellon University (CMU), and organized by the GenLaw Center (A. Feder Cooper, Katherine Lee, James Grimmelmann), Carnegie Mellon's K&L Gates Initiative (Hoda Heidari), the Georgetown Law Center (Paul Ohm), and the Center for Democracy and Technology (Alexandra Reeve Givens, Miranda Bogen). [[link](#)]

1st Workshop on Generative AI and Law (GenLaw '23). Held at *ICML 2023* in Honolulu, Hawai'i, 26% acceptance rate. Lead co-organizer with Katherine Lee; co-organized with Dr. Niloofar Mireshghallah, Madiha Zahrah Choksi, Prof. James Grimmelmann, Prof. David Mimno, and Dr. Deep Ganguli [[link](#)]

Selected Recent Talks

Invited participant, Symposium on AI and Law, Harvard Law School, October 24, 2025 (Cambridge, MA).

"What Copyright Can Learn from Memorization Measurements of Language Models."

Invited talk, American National Standards Institute (ANSI), May 11, 2026 (Virtual).

Invited speaker, Workshop & Lecture Series on the Law & Economics of Innovation, ETH Zürich, October 28–29, 2025 (Zürich, Switzerland).

Invited speaker, Information Technology and Innovation Foundation, August 2025 (Virtual).

Invited speaker, [Workshop on the Impact of Memorization on Trustworthy Foundation Models](#), *International Conference on Machine Learning*, July 2025 (Vancouver, Canada).

Invited panelist, [Workshop on Machine Unlearning for Generative AI](#), *International Conference on Machine Learning*, July 2025 (Vancouver, Canada).

AI Governance Tutorial (with Katherine Lee, 28th Annual BTLJ-BCLT Spring Symposium: AI Governance at the Crossroads, February 27, 2025 (Berkeley, CA).

"Machine Unlearning Doesn't Do What You Think: Lessons for Generative AI Policy, Research, and Practice."

Future of Privacy Forum, Technologist Roundtable for Policymakers, July 2025 (Virtual).

Invited talk, NYU Law Innovation Policy Colloquium, April 3, 2025 (New York, NY).